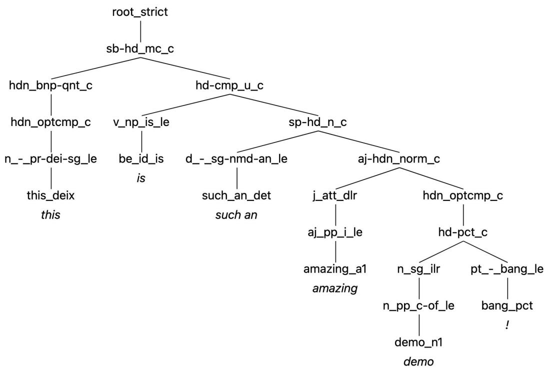
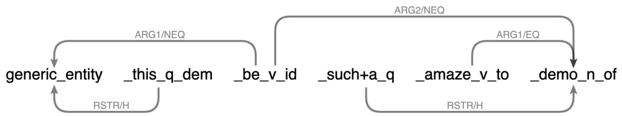
DELPH-IN Summit 2026

DELPH-IN Grammarium

Luis Morgado da Costa

VU VRIJE UNIVERSITEIT AMSTERDAM





TOP h0
INDEX e2
RELS ⟨ [generic_entity⟨0:4⟩ LBL h4 ARG0 x3], [_this_q_dem⟨0:4⟩ LBL h5 ARG0 x3 RSTR h6 BODY h7], [_be_v_id⟨5:7⟩ LBL h1 ARG0 e2 ARG1 x3 ARG2 x8], [_such+a_q⟨8:15⟩ LBL h9 ARG0 x8 RSTR h10 BODY h11], [_amaze_v_to⟨16:23⟩ LBL h12 ARG0 e13 ARG1 x8 ARG2 i14], [_demo_n_of⟨24:28⟩ LBL h12 ARG0 x8 ARG1 i15] ⟩
HCONS ⟨ [qeq HARG h0 LARG h1], [qeq HARG h6 LARG h4], [qeq HARG h10 LARG h12] ⟩

Summary

- Summary of main changes
- A quick DEMO
- Some open questions
- A look at the wishlist/Plans for the future

DELPH-IN Grammarium

- Web-based toolkit for grammar development
- PyDelphin, DELPH-IN Viz libraries, ACE
- Grammar Update/Compile (from GitHub)
    - Including through commit hashes/branches
- YY-input support (mainly for SRG)

New this year:

- Minor aesthetic improvements
- Docstrings on trees
- Profiles & Regression testing (are back and improved!)

DELPH-IN Grammarium

Type your sentence here...

Mandarin Resource Grammar [mrg.dat] | 1 Parse | Normal Input

Parse Sentence

Summary: MRG is a mid-sized grammar, with targeted coverage. Much of its development has been centered around educational data, which means it is able to provide adequate analyses for most of the sentence patterns found in early to intermediate educational materials. From a linguistic point of view, MRG offers in-depth analyses of interesting and well-studied linguistic phenomena including: serial-verbs, 的 (DE) constructions, 把 (BA) constructions, interactions between aspect and negation, sentence final particles, and interrogatives.

Input Notes: MRG expects its input to be segmented at the word level. This means that words must be separated by spaces (see some of the examples below). Some of our analyses expect input to be tokenized differently than most Mandarin Chinese word segmentation tools (e.g., 这个 should be tokenized as two tokens 这 and 个). To the best of its ability, the grammar uses REPP (internal preprocessing) to help further segment the input (e.g., if 这个 is provided as a single token, it will be internally split as two tokens).

Examples: ?

- 我 正在 看 电视 呢 。
- 这 个 商场 很 大 。
- 我们 还 要 三 碗 汤 。
- 你 会 不 会 开车 ?
- 语言 学院 有 多少 个 系 ?

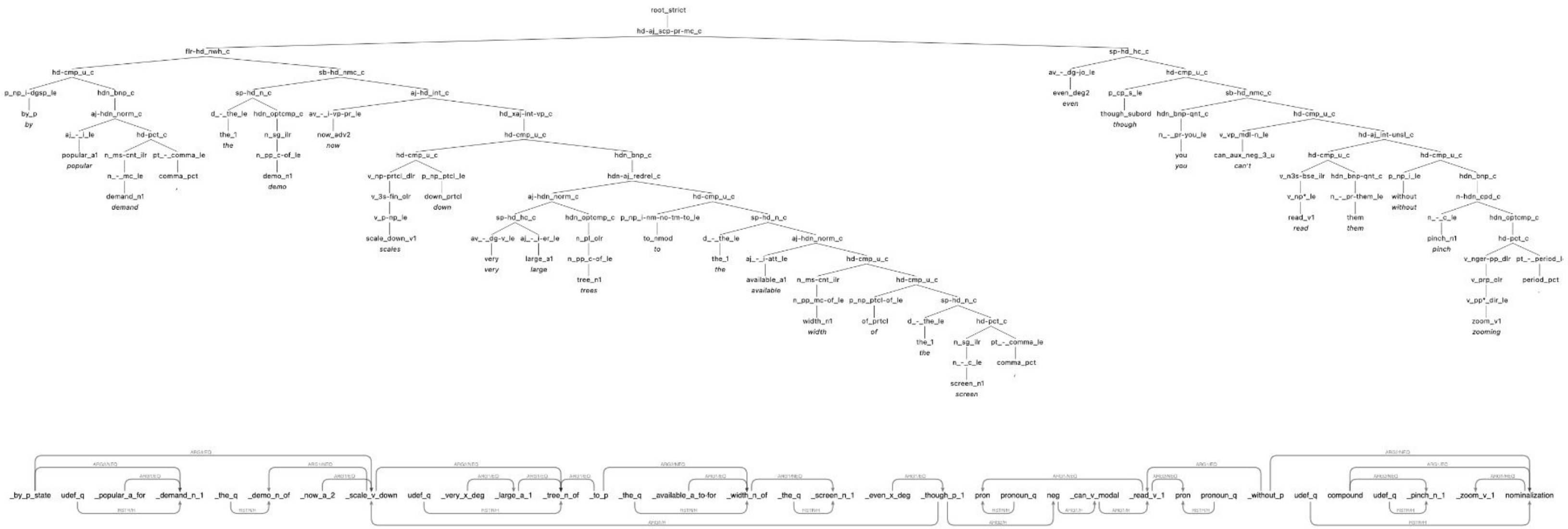
DEMO

By popular demand: Miniscule Trees

1 Parse Result

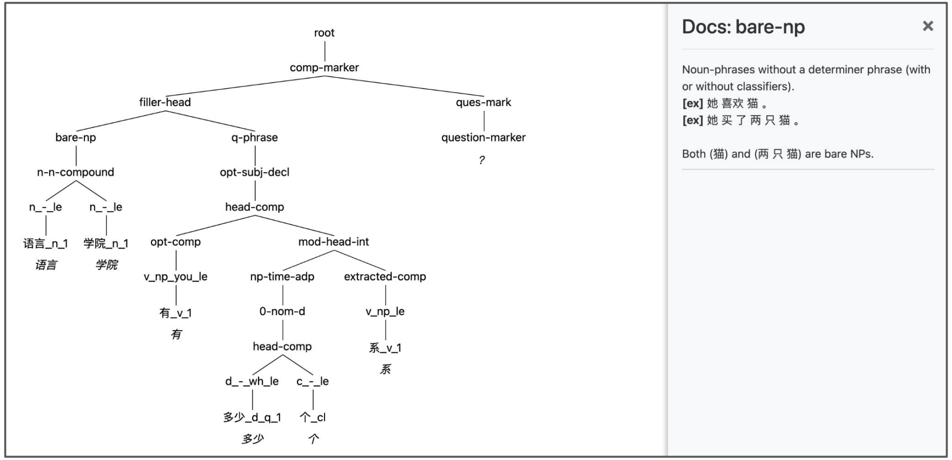
❝ By popular demand, the demo now scales down very large trees to the available width of the screen, even though you can't read them without pinch zooming. ❞

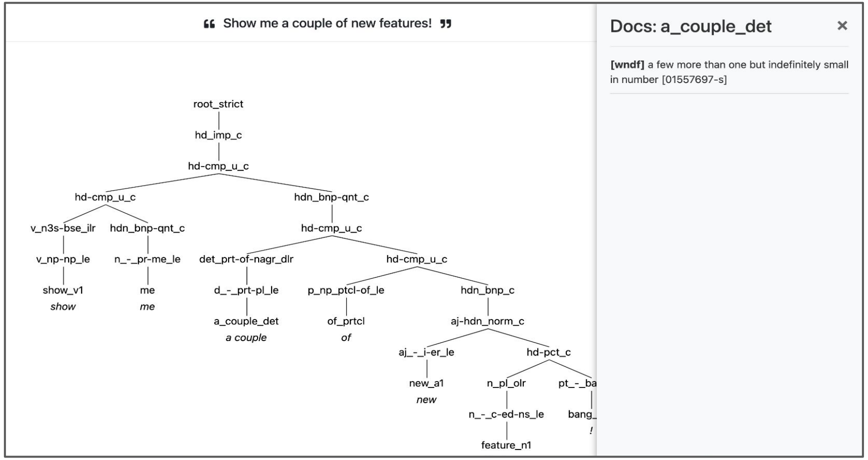
1



Very large trees now shrink down to available screen width… And surface strings!

DocStrings on Parsing Trees





- This was available in the FFTB… and now also in the online parser
- Only the ERG have MRG/ZHONG have **jsondocs**
- Currently, these **jsondocs** need to be hosted online (not sure if ideal)

  e.g. https://delph-in.github.io/erg/statics/erg-docs.json

DocStrings JSON (Jsondocs)

- It is a work in progress...
- Simple structure
  - Text, Type/Instance, Parents
  - Parents (not in use yet)
- My docstrings have full HTML tags (I know other don't like this)
- I think we need to agree on the fields of this JSON, but let each person use the format and convert their docstrings as they want it... [SIG?]


```
    {
        "docstring": ";; Bare NP phrase.  Consider modifying the PRED value of the quantifier <br>;; relation introduced to match the semantic effect 
of bare NPs in your <br>;; language.<br><br>;; MATRIX REDEFINITION<br>;; Note: we removed the SPEC <> constraint<br>;;; DPF - motivate? - shouldn't thi
e head at least propagate SPEC to mother?<br><br>FIXME LMC&FCB tried to made MC - to stop punctuation-end to join; but it is somehow breaking adjectiv
es to pick up normal nouns... <br><br>Needs to pass some important features, for example, NONSUBJ (whether something was marked as impossible to becom
e a subject)",
        "name": "basic-bare-np-phrase",
        "parents": [
            "head-only"
        ]
    },
    {
        "docstring": "This rule transforms a common noun without a determiner to be NP.<br>This results on a NP marked as [BARE +].<br><br>We added c-
cont-1 to propagate cfrom and cto to the to the mrs.<br>It inherits from 'c-cont-1' because we only introduce one (hence 1) <br>predicate in the mrs.<
br><br><br>This is not a 'headed rule', since it changes the valency. So we must carefully pass up everything that might be relevant. For example, LOC
. [FIX] I believe a few other things might be missing.<br><br><br><b>[ex]</b>我 吃 苹果 。<br>(I eat apples.)<br><br>Added [INTERJECTED #interjected] t
o copying. This helps block modification (with interjections) of NPs if they are going through further process. At the very least, this blocks certain
 phrases from being pumped by np-time-adp.",
        "name": "bare-np-phrase",
        "parents": [
            "basic-bare-np-phrase",
            "c-cont-1"
        ]
    },
    {
        "docstring": "[ EMPTY - ] on child node to prevent temporal phrases like 昨天 from being converted from ADP to NP again.<br>And [ EMPTY + ] on 
 parent node to signal this NP has no head noun.<br><br><br>LMC: Why was this forced as [INDEX.SORT nontime-s] ?",
        "name": "basic-bare-nominal-phrase",
        "parents": [
            "unary-phrase"
        ]
    },
    {
        "docstring": "pushing DE construction to be NP, including relativing DE and associative DE. DeP should have its COMP fulfilled, thus avoid nom
inalizing DE alone.<br><b>[ex]</b>我 喜欢 很 红 的<br><b>[ex]</b>我 喜欢 他 写 的<br><b>[ex]</b>我 喜欢 吃 的<br><b>[ex]</b>我 的 哭 了<br><b>[ex]</b>我__喜
欢__那__本__红__的<br><b>[ex]</b>我__喜欢__的__是__那__本__红__的<br>Ideally for ADJ/V + DE, only the slashed DeP should be pushed up to be NP. But adding in
 this constraint would prevent the rule<br>from being applied for possessive NP+DE. To be fixed later.<br><br><br>LMC/BOND: 的_comp is allowing non ex
tracted subjs (opt-subj-decl), that is weird (it should be blocked on the DE). <br>We want 0-nom-p would need to block the associative DE (not sure ho
w yet).<br><br><br>This 0-nom-p phrases were also allowed to be pumped into locative adjuncts.<br>As such, we want to add SYNSEM.LOCAL.CAT.HEAD.LOC na
-or--  to the NPs produced by this 的 in case of 'missing NPs'.<br><br><br>[SORT nontime-s] to block these pumped postposition phrases into time adjuc
ts.",
        "name": "bare-nominal-postp-phrase",
        "parents": [
            "basic-bare-nominal-phrase"
        ]
    },
```


Note: LKB-FOS: Docstrings in irules are not accepted...

Grammar Configuration

- Grammars are configured through a set of descriptors
- Currently they are set as a dictionary >> soon this will be a JSON file per grammar
- Some are optional
- A couple (repo_name & grammar file) <u>NEED</u> to be unique unique

```
GRAMMARS_CONFIG["mrg"]["repo_name"] = "mrg"
GRAMMARS_CONFIG["mrg"]["github_url"] = "https://github.com/lmorgadodacosta/MRG"
GRAMMARS_CONFIG["mrg"]["grammar_file"] = "mrg.dat"
GRAMMARS_CONFIG["mrg"]["grammar_title"] = "Mandarin Resource Grammar"
GRAMMARS_CONFIG["mrg"]["language"] = "Mandarin Chinese"
GRAMMARS_CONFIG["mrg"]["path_to_ace_config"] = "ace/config.tdl"
GRAMMARS_CONFIG["mrg"]["path_to_skeletons"] = "tsdb/skeletons"
GRAMMARS_CONFIG["mrg"]["path_to_gold_profiles"] = "tsdb/gold"
GRAMMARS_CONFIG["mrg"]["url_to_jsondocs"] = "https://lmorgadodacosta.github.io/MRG/static/docs.json"
GRAMMARS_CONFIG["mrg"]["description"] = ("MRG is a mid-sized grammar, with targeted coverage.  Much of its "
                                         "development has been centered around educational data, which means it is "
                                         "able to provide adequate analyses for most  of the sentence patterns "
                                         "found in early to intermediate educational materials.  From a linguistic "
                                         "point of view, MRG  offers in-depth analyses of interesting and "
                                         "well-studied linguistic phenomena including: serial-verbs, 的 (DE) "
                                         "constructions, 把 (BA) constructions, interactions between aspect and "
                                         "negation, sentence final particles, and  interrogatives.")
GRAMMARS_CONFIG["mrg"]["input_notes"] = ("MRG expects its input to be segmented at the word level. This means that "
                                         "words must be separated by spaces (see some of the examples below). Some "
                                         "of our analyses expect input to be tokenized differently than most  "
                                         "Mandarin Chinese word segmentation tools (e.g., 这个 should be tokenized "
                                         "as two tokens 这 and 个). To the best of its ability, the grammar uses "
                                         "REPP (internal preprocessing) to help further segment the input (e.g., "
                                         "if  这个 is provided as a single token, it will be internally split as "
                                         "two tokens).")
GRAMMARS_CONFIG["mrg"]["examples"] = ["我 正在 看 电视 呢 。",
                                      "这 个 商场 很 大 。",
                                      "我们 还 要 三 碗 汤 。",
                                      "你 会 不 会 开车 ？",
                                      "语言 学院 有 多少 个 系 ？",
                                      ]
```

Inspecting & Parsing Profiles

Inspecting Profile: mrs (ERG 2023)

?

Show 5 entries

Showing 1 to 5 of 107 entries

Previous 1 2 3 4 5 ... 22 Next

i-id	i-wf	readings	i-input	i-comment	
11	1		It rained.	Det regnet.	PARSE
21	1		Abrams barked.	Abrams bjeffet.	PARSE
31	1		The window opened.	Vinduet åpnet seg.	PARSE
41	1		Abrams chased Browne.	Abrams jaget Browne.	PARSE
51	1		Abrams handed Browne the cigarette.	Abrams rakte Browne sigaretten.	PARSE
?	?	?	?	?	

ERG 2023 [erg-latest.dat]

1 Parse

Parse Full Skeleton

Inspecting & Parsing Profiles

What parts of the profile do you think would be useful to see in these views?

Show 5 entries

Showing 1 to 5 of 107 entries — Previous 1 2 3 4 5 … 22 Next

		readings					
id	wf	old	new	Δ	length	input	
11	1	0	1	1	2	It rained.	PARSE
21	1	0	1	1	2	Abrams barked.	PARSE
31	1	0	2	2	3	The window opened.	PARSE
41	1	0	2	2	3	Abrams chased Browne.	PARSE
51	1	0	3	3	5	Abrams handed Browne the cigarette.	PARSE
?	?	?	?	?	?	?	

ERG 2023 [erg-latest.dat] — 1 Parse

Inspecting Profile: mrs (ERG 2023)

Show 5 entries

Showing 1 to 5 of 107 entries — Previous 1 2 3 4 5 … 22 Next

i-id	i-wf	readings	i-input	i-comment	
11	1		It rained.	Det regnet.	PARSE
21	1		Abrams barked.	Abrams bjeffet.	PARSE
31	1		The window opened.	Vinduet åpnet seg.	PARSE
41	1		Abrams chased Browne.	Abrams jaget Browne.	PARSE
51	1		Abrams handed Browne the cigarette.	Abrams rakte Browne sigaretten.	PARSE
?	?	?	?	?	

ERG 2023 [erg-latest.dat] — 1 Parse

Parse Full Skeleton

Profile inspection and Regression tests can (and are) a bit different…

Running and Comparing Regression tests

Regression Results

	Old Profile	New Profile	Δ
Profile Path	delphin/regressions/mrs/	/Users/lmc/Documents/work-projects/itell/delphin/regressions/2026-07-25_22:52:37___mrs___erg-latest.dat/	-
Number of Sentences	107	107	0
Number of Parsed Sentences	0	107	107
% of Parsed Sentences	0.00%	100.00%	100.00%
Avg. Parsed Sentence Ambiguity	-	6.24	-
Avg. Sentence Length (all)	4.47	4.47	0.00
Avg. Parsed Sentence Length	0.00	4.47	4.47

Quick Summary

Many things missing (what would be nice to show here?)

Show 5 entries

Showing 1 to 5 of 107 entries

Previous 1 2 3 4 5 ... 22 Next

		readings						
id	wf	old	new	Δ	length	input		
11	1	0	1	1	2	It rained.		PARSE
21	1	0	1	1	2	Abrams barked.		PARSE
31	1	0	2	2	3	The window opened.		PARSE
41	1	0	2	2	3	Abrams chased Browne.		PARSE
51	1	0	3	3	5	Abrams handed Browne the cigarette.		PARSE
?	?	?	?	?	?	?		

ERG 2023 [erg-latest.dat]

1 Parse

Inspect the profile in relation to a previous test

What comparisons would you want to see?

To the Wishlist!

https://docs.google.com/document/d/1eaiCseAypsvFFb_MMDx9wwLHLPXi21mOhuc3BAMuTIM/edit?usp=sharing (view only)

Thank you!

VRIJE UNIVERSITEIT AMSTERDAM